

Article

Core–Periphery Organization and Spatial Heterogeneity in *Pseudopus apodus* (Anguidae) Across Its Western Palearctic Range

Mehmet Kürşat Şahin ^{1,2,*} , Azra Topal ³  and Muammer Kurnaz ⁴ ¹ Biology Department, Faculty of Sciences, Hacettepe University, 06800 Ankara, Türkiye² Biological Diversity Research and Application Center, Hacettepe University, 06800 Ankara, Türkiye³ Molecular Biology and Genetics Department, Faculty of Sciences, Ondokuz Mayıs University, 55200 Samsun, Türkiye; azratopalbiologus@gmail.com⁴ Department of Medical Services and Techniques, Kelkit Sema Dogan Vocational School of Health Services, Gümüşhane University, 29600 Gümüşhane, Türkiye; muammerkurnazz@gmail.com

* Correspondence: kursat.sahin@hacettepe.edu.tr

Abstract

Understanding the internal spatial structure of widely distributed species is fundamental for biogeographic theory and conservation practice, yet such structure is often masked by extent-based range metrics. We investigated the spatial organisation of *Pseudopus apodus* (Pallas, 1775), the largest limbless lizard of the Western Palearctic, using 3967 occurrence records spanning 1843–2025. Spatial point pattern analysis revealed a pronounced deviation from complete spatial randomness (Clark–Evans $R = 0.105$, $p < 0.001$), with strong fine-scale clustering. Kernel density estimation identified a clear core–periphery organisation: high-density core areas accounted for 30% of records but occupied only 22% of the total extent of occurrence (EOO). The discrepancy between EOO (~8.9 million km²) and area of occupancy (AOO; ~8944 km²) spanned three orders of magnitude, emphasising that only a small fraction of the species' geographic envelope is actively occupied. Spatial heterogeneity was high (coefficient of variation ≈ 0.99), and core and peripheral occurrences were significantly segregated along both latitudinal and longitudinal gradients. The proportion of core records showed a weak positive temporal trend most plausibly attributable to sampling effort, particularly the recent expansion of citizen-science contributions, rather than to ecological processes. These findings demonstrate that *P. apodus* exhibits a compact spatial core embedded within a broad, sparsely occupied periphery, underscoring the limitations of EOO-based metrics in the conservation assessment of widely distributed reptiles. We emphasise that this structure is characterised using the density of occurrence records and therefore describes the observed spatial organisation of the available data rather than directly measured population density; ecological interpretations are accordingly framed as hypotheses requiring independent validation.

Keywords: sheltopusik; biogeography; core–periphery hypothesis; extent of occurrence; area of occupancy; kernel density estimation; reptile distribution



Academic Editor: Michael Wink and Miguel Ferrer

Received: 14 May 2026

Revised: 8 June 2026

Accepted: 12 June 2026

Published: 16 June 2026

Copyright: © 2026 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

1. Introduction

Understanding the spatial structure of species distributions is a central goal in ecology and biogeography, providing insights into the processes that govern species persistence, range limits, and responses to environmental change [1,2]. Species are rarely

distributed uniformly across their geographic ranges; instead, they exhibit spatial heterogeneity driven by environmental gradients, habitat availability, dispersal constraints, and historical factors [3,4]. Identifying and quantifying such patterns is therefore critical for both theoretical ecology and applied conservation, particularly under accelerating environmental change.

One of the most widely discussed frameworks describing spatial variation within species ranges is the core–periphery hypothesis, which posits that population density and individual performance are highest near the centre of a species’ range and decline toward its margins [1,5]. Core regions are typically associated with optimal environmental conditions and stable populations, whereas peripheral regions often consist of fragmented, lower-density populations that may be more vulnerable to environmental fluctuations [6,7]. Despite its intuitive appeal, empirical support for the core–periphery hypothesis remains mixed, and its generality continues to be debated [5,8]. This highlights the need for robust quantitative approaches capable of revealing internal spatial structure within widely distributed species.

In parallel, conservation biology has emphasised the importance of distinguishing between the extent of occurrence (EOO) and the area of occupancy (AOO) when assessing species’ spatial status [9]. While EOO describes the overall geographic extent of a species, it often includes large areas of unsuitable habitat and may therefore overestimate the true spatial footprint of viable populations [10]. AOO provides a more realistic measure of the area actually occupied. The discrepancy between these metrics can be substantial and has important implications for both Red List assessments and management strategies [9,11].

Recent advances in spatial ecology have enabled the analysis of species distributions using point pattern approaches, which treat occurrence records as realisations of an underlying spatial process [12]. Such methods allow for the detection of clustering, spatial heterogeneity, and scale-dependent structure without requiring explicit environmental predictors. In particular, nearest-neighbour statistics and Ripley’s K function have been widely used to quantify deviations from complete spatial randomness (CSR) and to identify aggregation patterns across spatial scales [13,14]. When combined with kernel density estimation, these approaches can reveal high-intensity regions within a species’ range and provide a basis for defining core areas in a data-driven manner [15,16].

Reptiles represent particularly suitable models for investigating internal spatial structure because of their strong dependence on microhabitat conditions, thermoregulatory constraints, and limited dispersal [17,18]. These traits often translate into patchy distributions and localised population clusters across landscapes. *Pseudopus apodus* (Pallas, 1775), the sheltopusik or European glass lizard, is one of the most widely distributed reptiles of the Western Palearctic and offers a strong model for examining how broad geographic ranges are internally structured. As the largest limbless lizard of the region, reaching total lengths exceeding 1.3 m, the species inhabits Mediterranean shrublands, rocky slopes, grasslands, steppe-mosaic ecosystems, and abandoned cultivation. Its range extends across an exceptionally wide longitudinal band stretching from southern Europe and the Balkans through Anatolia and the Caucasus to Central Asia [19,20]. Two subspecies are currently recognised: *Pseudopus apodus thracicus* in the Balkan Peninsula and *Pseudopus apodus apodus* across Anatolia, the Caucasus, the Levant, Iran, and Central Asia [19,21]. The species is currently listed as Least Concern by the IUCN, although habitat loss, agricultural intensification, road mortality, and persecution stemming from misidentification as a snake generate localised population declines [22].

Despite its wide distribution and biogeographic interest, the internal spatial organisation of *P. apodus* across its range has not been quantitatively assessed. In particular, it remains unclear whether the species exhibits a clear core–periphery structure, how strongly

its distribution deviates from spatial randomness, and how its effective occupied area compares to its overall geographic extent. The increasing availability of occurrence data over time also raises important questions about the relative roles of sampling effort versus genuine ecological processes in shaping observed distribution patterns [23,24].

In this study, we use a comprehensive dataset of occurrence records spanning nearly two centuries to address these gaps. Specifically, we aim to (i) evaluate the degree of spatial aggregation and deviation from randomness; (ii) identify and quantify core and peripheral regions within the distribution; (iii) assess the relationship between EOO and AOO across both core and peripheral subsets; and (iv) examine temporal changes in the spatial structure of occurrences while explicitly accounting for variation in sampling effort. By integrating point pattern analysis, kernel density estimation, and standard range metrics, this study provides the first comprehensive quantitative characterisation of the internal spatial structure of *P. apodus* and contributes a fresh perspective to the broader debate on internal range organisation in widely distributed reptiles.

2. Materials and Methods

2.1. Study Species and Occurrence Data

Pseudopus apodus is a large, diurnal anguid lizard endemic to the Western Palearctic. The species occupies thermophilous open habitats, including Mediterranean shrublands, dry rocky slopes, grasslands, agricultural margins, riparian zones, and steppe mosaics, typically below 1500 m elevation [19,20]. Its diet is dominated by terrestrial gastropods, large arthropods, small vertebrates, and occasionally fruit [19,21].

Occurrence records of *P. apodus* were compiled from a comprehensive dataset spanning 1843–2025 (Figure 1), drawing primarily from publicly available biodiversity repositories, including the Global Biodiversity Information Facility (GBIF; <https://www.gbif.org>) and iNaturalist research-grade observations, supplemented by published distributional atlases of European and Western Palearctic herpetofauna [19,20,25]. Each record included geographic coordinates (latitude and longitude), observation date, and country information. An initial pool of 4685 georeferenced records was retrieved. Records lacking geographic coordinates were excluded prior to analysis. Duplicate records sharing identical coordinates were removed to avoid spatial bias associated with repeated observations at the same locality [26]. Spatially isolated records exceeding 200 km from their nearest neighbour were cross-checked against published distribution maps and regional atlases [19,20] to confirm consistency with the known biogeographic range of the species and to filter potential georeferencing errors. Observation dates were standardised using flexible date-parsing functions, and the year of each record was extracted for subsequent temporal analyses. After these sequential cleaning steps (removal of records without coordinates, removal of exact coordinate duplicates, and verification of spatial outliers), 3967 of the initial 4685 records were retained for analysis. The final dataset comprised 3967 unique occurrence records covering the full known geographic range of the species across the Balkans, Anatolia, the Caucasus, the Levant, Iran, and Central Asia [19,20]. The complete list of all 3967 occurrence records used in the analysis, including coordinates, observation year, country, and data source, is provided in full in the Supplementary Material (Table S1) to ensure reproducibility and to allow independent verification of the localities analysed.

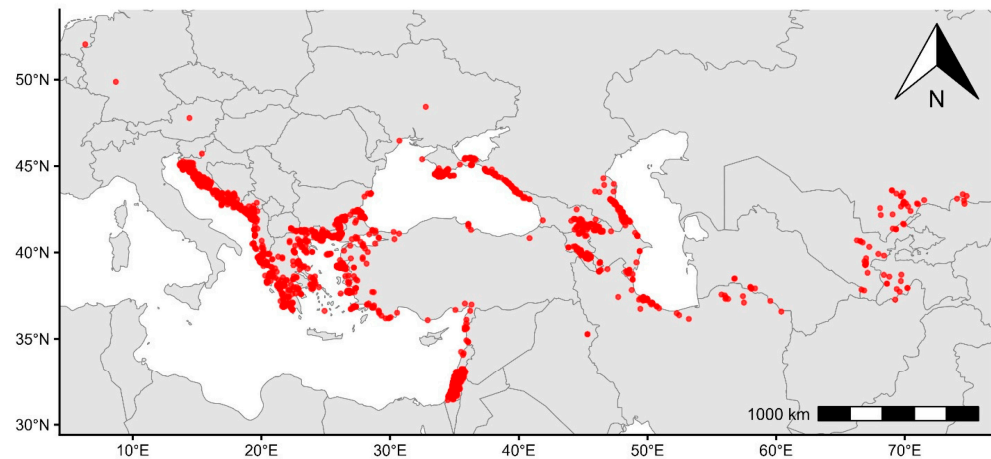


Figure 1. Spatial distribution of occurrence records of *Pseudopus apodus* across its Western Palearctic range (1843–2025), spanning the Balkans, Anatolia, the Caucasus, the Levant, Iran, and Central Asia.

2.2. Spatial Data Processing

All spatial analyses were conducted in R version 4.4 [27]. Occurrence data were converted to spatial objects using the *sf* package v.1.1 [28]. To enable accurate distance-based analyses, occurrence coordinates were projected into a metric coordinate reference system (UTM Zone 36N; EPSG:32636), which approximately bisects the longitudinal extent of the dataset.

Spatial point pattern analyses were performed using components of the *spatstat* family of packages, including *spatstat.geom* v.3.8, *spatstat.explore* v.3.8, and *spatstat.random* v.3.5 [12]. A spatial observation window was defined using the convex hull of occurrence points with an added 50 km buffer to reduce edge effects in spatial statistics [14].

2.3. Point Pattern Analysis

Spatial aggregation was quantified using the Clark–Evans nearest-neighbour index [13], which compares observed nearest-neighbour distances to those expected under complete spatial randomness (CSR). Statistical significance was assessed using a Z-test, with a Donnelly edge correction applied to mitigate boundary effects.

The full distribution of nearest-neighbour distances was further examined to characterise local spatial structure and to detect clustering versus dispersion patterns at fine spatial scales. Multi-scale spatial structure was evaluated using Ripley’s K function [14], with Monte Carlo simulations ($n = 99$ random realisations of CSR) used to generate confidence envelopes and assess scale-dependent deviations from CSR.

2.4. Kernel Density Estimation and Core–Periphery Classification

Spatial intensity of occurrence records was estimated using kernel density estimation (KDE), implemented in the *spatstat.explore* package [12]. A fixed bandwidth of 10 km was applied across the study area, selected to capture landscape-scale aggregation patterns rather than fine-scale individual distributions, given the broad geographic extent of the dataset and the macroecological focus of the analysis [15].

2.5. Core–Periphery Spatial Metrics

Convex hull polygons were used to quantify the extent of occurrence (EOO) for the entire dataset and for the core and peripheral subsets separately. EOO was calculated as the area of the minimum convex polygon enclosing the relevant set of occurrence points, following IUCN Red List guidelines [9].

Area of occupancy (AOO) was estimated using a 2×2 km grid, in line with IUCN protocols [9]. Grid cells intersecting at least one occurrence record were considered occupied. Separate AOO estimates were computed for core and peripheral subsets to compare the effective occupied area of each component.

A contraction ratio was calculated as the ratio between core EOO and total EOO, providing a quantitative measure of spatial concentration within the species' geographic range.

2.6. Spatial Heterogeneity and Distance Metrics

Spatial heterogeneity in occurrence intensity was quantified using the coefficient of variation (CV) of kernel density values across the study area [3]. Distances between all occurrence points and their nearest core point were calculated to assess the spatial separation between core and peripheral regions, providing a metric of the spatial isolation of peripheral occurrences across the range.

2.7. Geographic Gradient Analysis

Latitudinal and longitudinal coordinates were extracted for each occurrence point and compared between the core and peripheral subsets. Differences were assessed using the non-parametric Wilcoxon rank-sum test [29], which does not assume normality of the underlying distributions.

2.8. Temporal Analysis

Temporal dynamics were assessed by assigning each occurrence record to a calendar year based on parsed observation dates. The temporal trend analysis was restricted to the period from 1990 onwards, when annual sample sizes were sufficient to allow meaningful proportions to be calculated; in earlier years, sparse and irregular reporting produced extreme stochastic values (proportions of 0 or 1 driven by single records) that would have introduced disproportionate noise into the trend estimation. The proportion of core occurrences was calculated for each year within this period, and the temporal trend was assessed using Spearman's rank correlation. Because increases in occurrence reporting over recent decades may reflect changes in sampling effort rather than ecological processes alone [23,24], temporal results are interpreted with appropriate caution and contextualised against the documented growth of citizen-science contributions to herpetological databases.

2.9. Software and R Packages

All analyses were performed in R version 4.4 [27], using the following packages: sf [28] for spatial data handling, spatstat [12] for spatial point pattern analysis, dplyr v1.1.4 [30] for data processing, lubridate v1.9.5 [31] for date parsing, dbscan v1.2.5 [32] for density-based clustering, and spThin v0.2.0 [33] for sampling bias assessment.

3. Results

3.1. Spatial Aggregation and Point Pattern Structure

The spatial distribution of *Pseudopus apodus* occurrences exhibited a strong and highly significant deviation from complete spatial randomness. The Clark–Evans nearest-neighbour index yielded a corrected value of $R = 0.105$ ($p < 2.2 \times 10^{-16}$), indicating pronounced clustering across the study area.

Nearest-neighbour distance analysis further supported this pattern. Distances were strongly right-skewed, as evidenced by a median of approximately 623 m being much smaller than the mean of 3.2 km, with maximum distances exceeding 300 km. The histogram of nearest-neighbour distances (Figure 2) shows a high frequency of short distances, indicating strong local aggregation, combined with a long tail representing spatially isolated records.

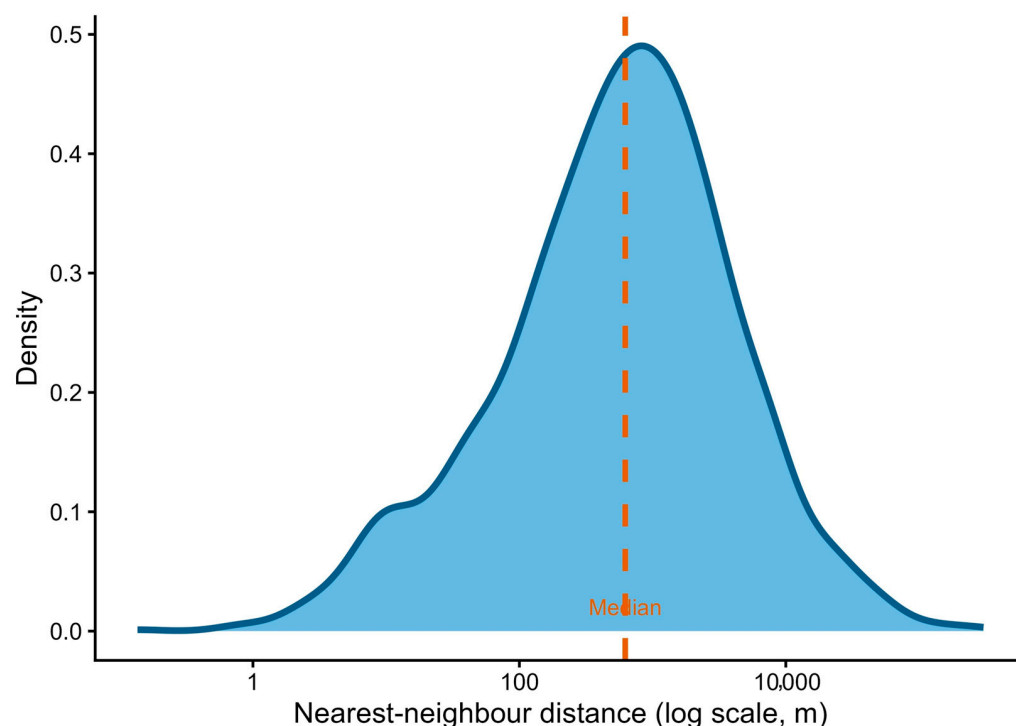


Figure 2. Distribution of nearest-neighbour distances among occurrence records of *Pseudopus apodus*. The density curve shows a strong concentration of short distances, indicating pronounced spatial clustering.

3.2. Spatial Heterogeneity and Density Structure

Kernel density estimation revealed a highly heterogeneous spatial intensity pattern. The distribution of density values (Figure 3) was strongly uneven, with a coefficient of variation of $CV = 0.99$, indicating substantial variation in occurrence intensity across the range.

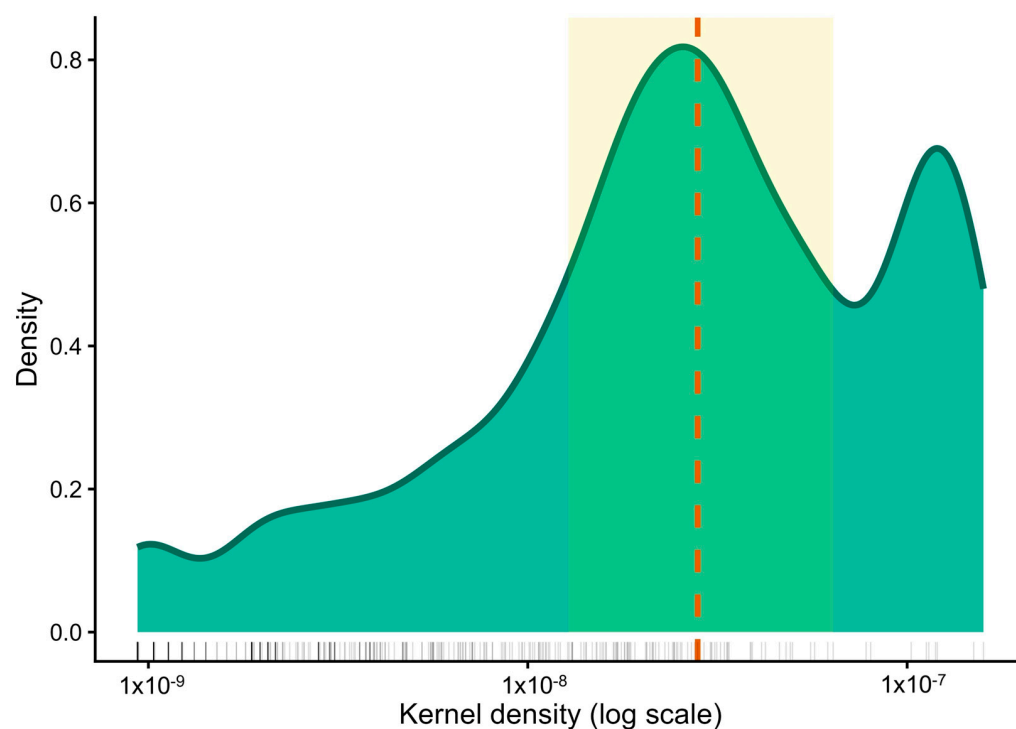


Figure 3. Distribution of kernel density values across occurrence records of *Pseudopus apodus*. The uneven distribution reflects substantial spatial heterogeneity in occurrence intensity.

When partitioned into core and peripheral components, density distributions differed markedly (Figure 4). Core occurrences were characterised by consistently higher density values, whereas peripheral records occupied lower-density regions, confirming clear spatial differentiation between these components.

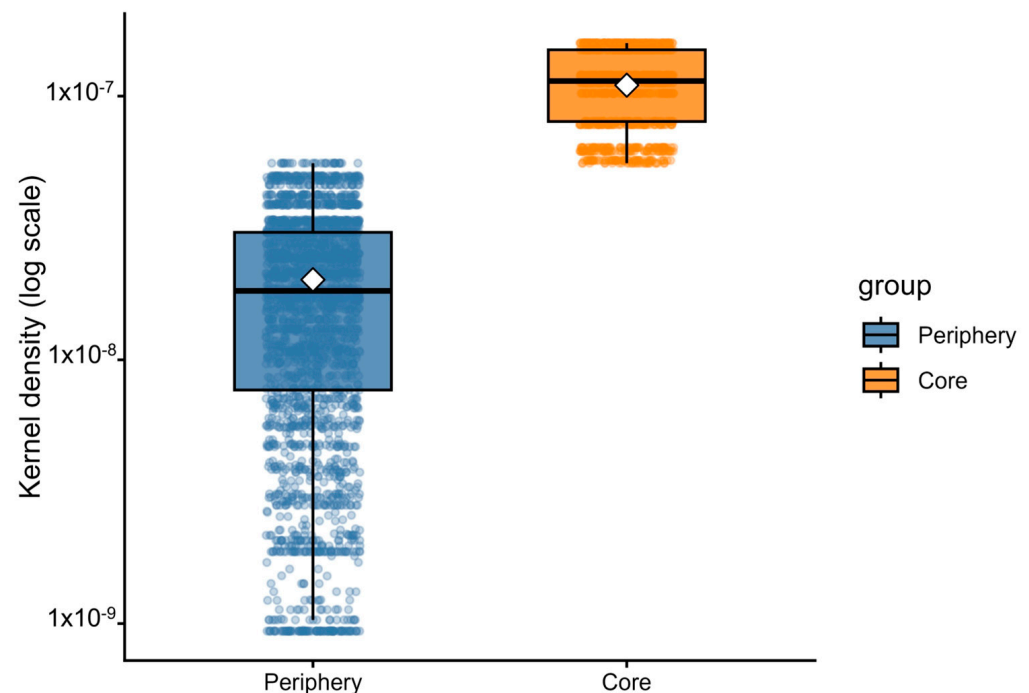


Figure 4. Comparison of kernel density values between core and peripheral occurrences of *Pseudopus apodus*. Boxplots illustrate higher density values in core areas relative to peripheral regions.

3.3. Core–Periphery Structure and Spatial Contraction

Using the upper 30% of kernel density values, 1190 records (30.0% of all observations) were classified as core, while 2777 records (70.0%) were classified as peripheral.

Convex hull analysis demonstrated a strong contraction between total and core distributions. The total extent of occurrence (EOO) was estimated at approximately 8.9 million km², whereas the core EOO covered only 1.93 million km², corresponding to 22% of the total range.

The area of occupancy (AOO) further emphasised this disparity. Total AOO was estimated at 8944 km² (corresponding to 2236 occupied 2 × 2 km grid cells), with approximately 1784 km² (≈20%) belonging to the core and 7204 km² (≈80%) to peripheral areas. This contrast highlights that a large proportion of the geographic range is sparsely occupied, while high-density occurrences are concentrated in a relatively small spatial subset. The spatial relationship between total distribution and core areas is illustrated in Figure 5, showing a compact high-density core embedded within a much broader distribution envelope.

3.4. Distance-to-Core Structure

Distances from all occurrences to the nearest core point revealed substantial spatial separation. The mean distance to the core was approximately 190 km, with maximum values exceeding 2300 km (Figure 6). The distribution of these distances indicates that peripheral occurrences are widely dispersed relative to the core, reinforcing the presence of a spatially structured system.

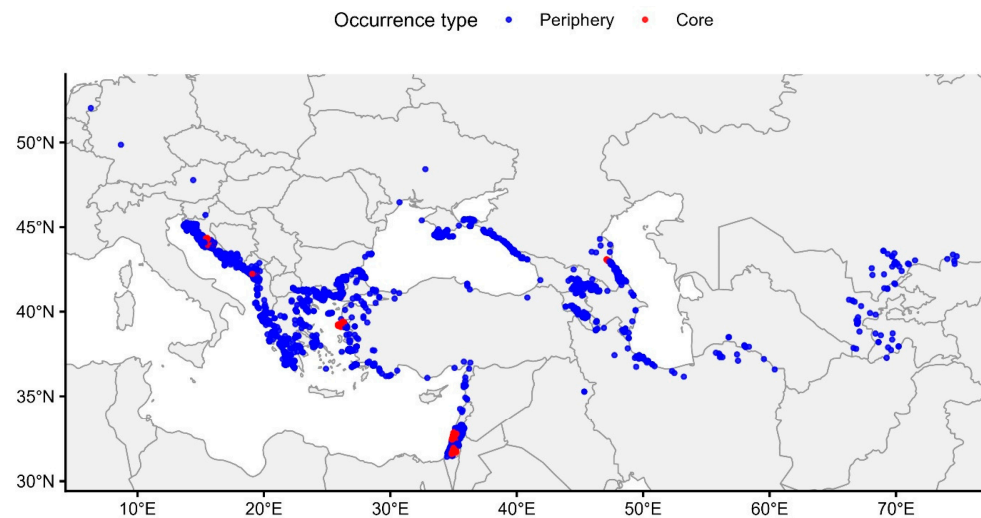


Figure 5. Spatial distribution of core and peripheral occurrences of *Pseudopus apodus*. Core occurrences (red) represent the upper 30% of kernel density values, while peripheral occurrences (blue) correspond to lower-density regions.

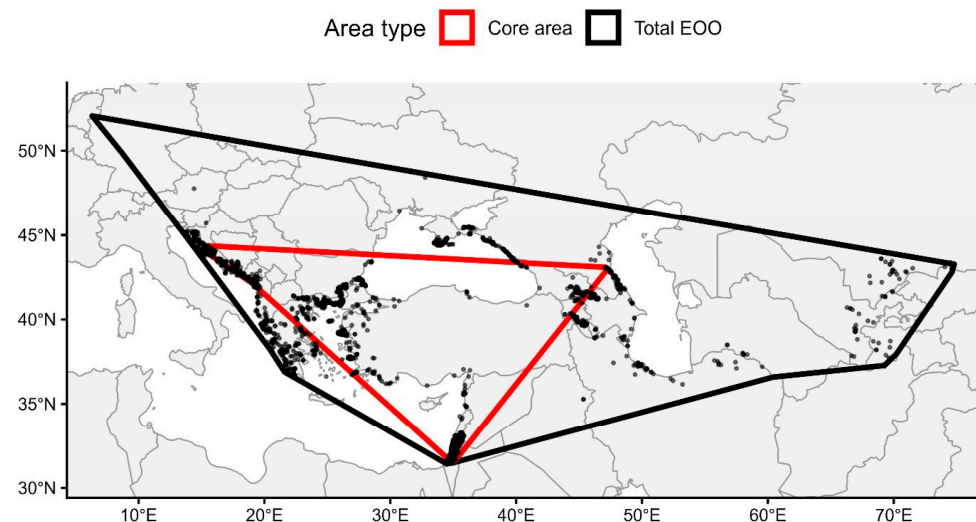


Figure 6. Comparison between the total extent of occurrence (EOO; black polygon) and the core area (red polygon) of *Pseudopus apodus*. The core region occupies a substantially smaller portion of the overall geographic range.

3.5. Geographic Gradients

Significant differences were observed between core and peripheral occurrences along both latitudinal and longitudinal gradients. Latitude differed strongly between groups (Wilcoxon rank-sum test, $p < 2.2 \times 10^{-16}$), and longitude also showed a significant difference ($p = 2.23 \times 10^{-5}$). Histograms of latitude and longitude distributions (Figure 7) illustrate this spatial segregation, indicating that core and peripheral occurrences occupy distinct geographic positions within the overall range.

3.6. Temporal Dynamics of Occurrence Records

The number of occurrence records increased markedly over time, particularly after 2000, reflecting a substantial rise in sampling intensity (Figure 8). Within the analysed period (1990–2025), the yearly proportion of core occurrences exhibited substantial inter-annual variability, including extreme values during years with very few records, alongside a more stable plateau in recent decades (Figure 9). A weak but statistically significant positive monotonic

trend was detected by rank correlation (Spearman $\rho = +0.29$, $p = 0.019$). It is worth noting that the local non-parametric smoothing displayed in Figure 9 reflects pronounced fluctuations during the early 1990s, when very small annual sample sizes generated extreme proportions, and should not be interpreted as a continuous decline; the rank-based correlation provides a more robust summary of the global temporal pattern across the time series. Given that this signal coincides with the rapid expansion of citizen-science platforms across the Western Palearctic [23,24], the trend most plausibly reflects an accumulation of casual observations in densely populated and easily accessible regions, which frequently coincide with high-density core areas, rather than a genuine ecological signal.

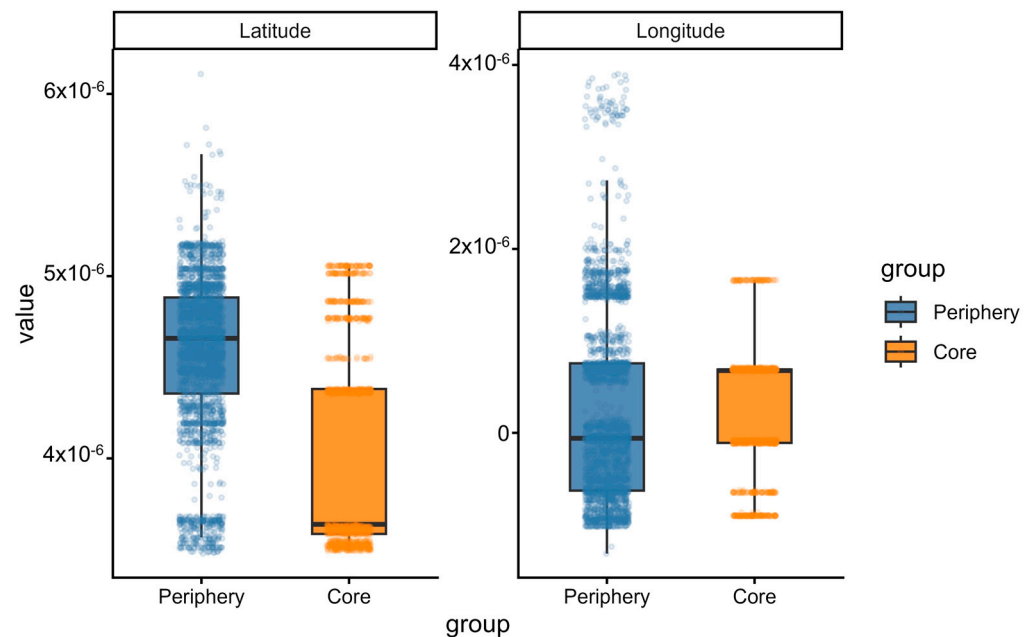


Figure 7. Latitudinal and longitudinal differences between core and peripheral occurrences of *Pseudopus apodus*. Boxplots represent the distribution of coordinate values, with jittered points showing individual observations.

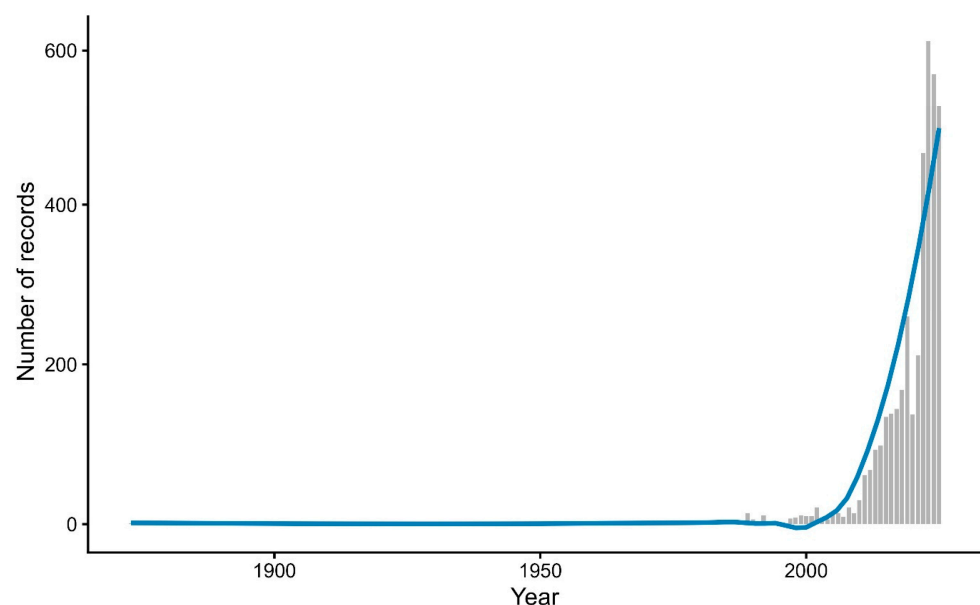


Figure 8. Temporal distribution of occurrence records of *Pseudopus apodus* across years. Bars represent the number of records per year, with a smoothed trend line illustrating changes in sampling intensity over time.

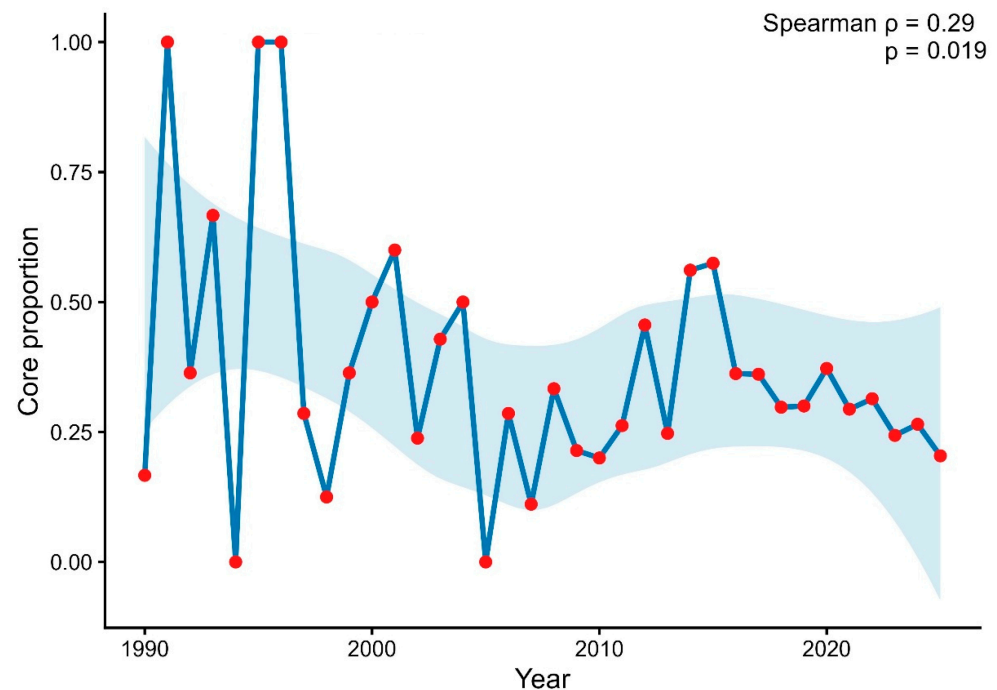


Figure 9. Temporal variation in the yearly proportion of core occurrences of *Pseudopus apodus* across the period 1990–2025. Red points indicate annual core proportions, the solid line connects sequential annual values, and the shaded ribbon represents a non-parametric local smoothing fit illustrating short-term variability. The pronounced fluctuations in the early 1990s are driven by very small annual sample sizes. Despite this volatility, a weak but statistically significant positive monotonic trend was detected by rank correlation (Spearman $\rho = +0.29$, $p = 0.019$).

4. Discussion

The present study reveals a strongly structured spatial distribution in *Pseudopus apodus*, characterised by pronounced clustering, a clear core–periphery organisation, and substantial discrepancies between the extent of occurrence (EOO) and area of occupancy (AOO). These patterns are consistent with a growing body of literature demonstrating that species distributions are rarely spatially homogeneous and instead reflect the integrated outcome of ecological, environmental, and historical processes [1,2].

The strong clustering detected in this study (Clark–Evans $R \ll 1$) indicates that occurrences are not randomly distributed but instead form localised aggregations. This pattern is typical of species whose distributions are driven by patchy habitat availability and environmental heterogeneity [34,35]. In reptiles, spatial clustering is often associated with microhabitat specificity, thermoregulatory requirements, and shelter availability [17]. *P. apodus* is particularly tied to mosaic landscapes that combine open foraging areas with structural refugia such as stone walls, rocky outcrops, dense scrub, and abandoned terraces [19,20], and its fine-scale distribution is therefore expected to track such heterogeneous landscape elements.

The nearest-neighbour distance distribution further supports this interpretation, showing a dominance of short inter-point distances combined with a long tail of isolated occurrences. Such hierarchical spatial structure, local clustering embedded within broader-scale dispersion, is commonly observed in species inhabiting fragmented or heterogeneous landscapes [4,36].

One of the most striking findings is the clear core–periphery structure, where a relatively small proportion of the range supports high-density occurrences. Core areas accounted for approximately 30% of records but only ~22% of the total EOO, indicating strong spatial contraction. Although our cores are defined statistically from occurrence

density rather than directly from environmental suitability, this pattern aligns with the classical core–periphery hypothesis, which predicts that population density and performance decline from the centre toward the margins of a species' range [1,5]. Core regions are typically associated with optimal environmental conditions, whereas peripheral areas often represent marginal habitats with lower suitability and more variable population dynamics [6].

For *P. apodus*, the spatial distribution of high-density core areas in the Balkans, western Anatolia, and the southern Caucasus is broadly consistent with regions widely identified as Pleistocene refugia for Western Palearctic herpetofauna [20,37], suggesting that contemporary core areas may partially preserve the imprint of long-term population persistence in climatically buffered zones. Importantly, these core areas also broadly correspond to regions of dense, continuous distribution depicted in published herpetological atlases [19,20], providing independent qualitative support for our density-based core classification. The finding that peripheral occurrences define the overall EOO further suggests that range limits are determined by sparse, possibly transient populations rather than continuous occupancy. This supports the idea that species ranges are often structured by a combination of stable core populations and dynamic peripheral populations [7,38].

The pronounced difference between EOO and AOO observed here, spanning three orders of magnitude, highlights the importance of distinguishing between geographic extent and actual occupancy. This discrepancy is widely recognised in conservation biology and forms the basis of IUCN Red List criteria [9].

EOO tends to overestimate the spatial extent of viable populations because it includes large areas of unsuitable or unoccupied habitat [10]. AOO provides a more realistic measure of the area actually used by the species. The results presented here clearly demonstrate that *P. apodus* occupies only a small fraction of its potential range, reinforcing the need to interpret EOO-based assessments with caution, particularly for widely distributed reptiles whose ranges traverse multiple biogeographic regions and major habitat boundaries.

The high coefficient of variation in density values ($CV \approx 1$) indicates a markedly heterogeneous spatial distribution. Such heterogeneity is a fundamental property of ecological systems and reflects variation in habitat quality, resource availability, and environmental constraints [3,39].

Spatial heterogeneity has important implications for population dynamics, including the formation of source–sink systems and metapopulation structures [40,41]. The large distances observed between peripheral occurrences and core areas further suggest that populations may be spatially fragmented and weakly connected, particularly at the edges of the range. As is typical for large anguillid lizards, *P. apodus* is generally regarded as long-lived and relatively sedentary, traits that may amplify the implications of spatial fragmentation for gene flow and demographic resilience at range margins.

The significant differences observed along latitudinal and longitudinal gradients indicate that core and peripheral occurrences occupy distinct geographic positions. Such spatial segregation is consistent with large-scale environmental gradients that influence reptile distributions, including climate, topography, and land cover [1,42].

In reptiles, latitudinal gradients often reflect thermal constraints and the energetic costs of thermoregulation, whereas longitudinal gradients across the Western Palearctic capture aridity and continentality gradients spanning Mediterranean, Pontic, and Irano-Turanian biomes [18]. The two recognised subspecies of *P. apodus*, namely *Pseudopus apodus thracicus* in the Balkans and *Pseudopus apodus apodus* across Anatolia, the Caucasus, and farther east [19,21], are likely to occupy partially distinct climatic envelopes given their broadly allopatric ranges, and the latitudinal–longitudinal segregation observed here is consistent with this biogeographic partitioning at the intraspecific level.

The temporal analysis revealed a marked increase in occurrence records in recent decades, accompanied by a weak positive monotonic trend in the proportion of core occurrences across the period 1990–2025. The most parsimonious interpretation of this pattern is that the rapid growth of citizen-science platforms, particularly iNaturalist and GBIF-mobilised observations after 2015 [23,24], has disproportionately concentrated new records in densely populated and easily accessible regions, which frequently overlap with the species' high-density core areas. The pronounced inter-annual variability of the time series, including the extreme fluctuations during the early 1990s when annual sample sizes were small, further illustrates the dominant role of sampling stochasticity in shaping the observed temporal signal. While genuine range dynamics in response to environmental change cannot be ruled out [43,44], the temporal signal observed here should not be interpreted as evidence of demographic expansion or contraction without independent verification through standardised monitoring.

Importantly, the persistence of a well-defined core throughout the temporal extent of the dataset indicates that the fundamental spatial structure of the species is robust and was not generated artefactually by the recent surge in opportunistic observations.

Although *P. apodus* is currently classified as Least Concern globally [22], its strongly heterogeneous distribution carries practical conservation implications. Reliance on EOO alone may substantially overestimate the effective distribution and security of the species, as most of the range is sparsely occupied. Core areas likely represent zones of higher habitat suitability and population stability and may therefore be of disproportionate importance for long-term persistence [6]. Peripheral populations, while important for range dynamics and adaptive capacity, may be more vulnerable to environmental fluctuations and local extinction [11], particularly in regions affected by agricultural intensification, road mortality, and active persecution due to the species' superficial resemblance to snakes [22].

More broadly, this study demonstrates that occurrence-only data, when analysed using spatial point pattern methods, can provide valuable insights into internal range structure without requiring complex distribution models. Such approaches offer a complementary perspective to traditional species distribution modelling and can help reveal fundamental spatial organisation within widely distributed reptile species.

Several limitations should be acknowledged. First, occurrence records derived from biodiversity databases are subject to spatial sampling bias, with effort concentrated in well-surveyed regions and underrepresentation in remote parts of the range, particularly in Iran and Central Asia [24,26]. Future analyses incorporating spatially explicit thinning at multiple radii would further test the robustness of core delineation under heterogeneous sampling intensity. Second, the use of UTM Zone 36N for distance and area calculations may introduce minor distortion at the eastern margins of the range; future analyses adopting equal-area projections (e.g., Lambert Azimuthal Equal Area) would refine quantitative estimates without altering the qualitative spatial structure documented here. Third, the convex hull-based EOO incorporates large areas of unsuitable habitat, including water bodies and high mountain ranges; α -hull or land-clipped approaches would yield more conservative EOO estimates. Fourth, the definition of core areas based on density thresholds is sensitive to bandwidth selection and threshold choice; sensitivity analyses across alternative bandwidths (e.g., 5, 10, 20 km) and core thresholds (e.g., 25%, 30%, 50%) would strengthen the generality of the results. Fifth, the use of occurrence-only data without environmental predictors limits direct inference of ecological core areas; integration with bioclimatic variables and habitat layers in future work would clarify the extent to which observed cores reflect ecological optima as opposed to sampling intensity. Finally, treating *P. apodus* as a single taxonomic unit may mask differences between subspecies; subspecies-

level analyses are a promising direction once verified subspecific assignments become available for occurrence records.

Beyond these specific points, a more fundamental interpretive caveat warrants explicit statement. Our analyses describe the spatial structure of occurrence records, not directly measured biological density. Because the dataset is assembled from opportunistic sources (GBIF, iNaturalist, and published atlases), the density of records reflects the joint outcome of true species presence and spatially uneven sampling effort, observer activity, and accessibility. The high-density “core” regions identified here should therefore be read, in the first instance, as regions of high record density rather than as confirmed zones of peak population density, and the possibility that they partly reflect historically well-collected and easily accessible localities—a form of survivorship and observer bias—cannot be excluded. For the same reason, complete spatial randomness (CSR) is used here only as a conventional statistical baseline against which to quantify the degree of aggregation; it is not proposed as a biologically realistic expectation for a species whose distribution is constrained by climate, habitat, topography, and political sampling boundaries, and the observed deviation from CSR is interpreted as a description of clustering intensity rather than as evidence of a specific ecological mechanism. We further acknowledge that defining the core from the upper tail of the kernel density surface and then showing that core records have higher density values is, by construction, internally consistent rather than an independent test of biological core–periphery structure; the convergence of these statistically defined cores with regions of dense, continuous distribution in published atlases [19,20] and with recognised Pleistocene refugia [37] provides only qualitative, not confirmatory, support. Accordingly, the ecological themes discussed above—habitat suitability, refugia, source–sink dynamics, gene flow, and demographic resilience—are presented as hypotheses consistent with the observed spatial pattern, not as findings demonstrated by these occurrence-only data, and their rigorous evaluation will require environmental predictors, abundance or demographic data, and standardised survey designs that fall outside the scope of the present descriptive study.

5. Conclusions

This study provides the first comprehensive quantitative characterisation of the internal spatial structure of *Pseudopus apodus* across its entire Western Palearctic range. The species exhibits pronounced spatial clustering, a clear core–periphery organisation, and a striking three-orders-of-magnitude discrepancy between EOO and AOO. This organisation is defined using the density of occurrence records and is therefore reported as a robust description of the observed spatial data rather than as a directly measured population-density pattern. Core areas comprise only ~22% of the total range yet contain a disproportionate share of high-density records, while peripheral occurrences extend across vast but sparsely occupied areas. The geographic segregation of core and peripheral occurrences is consistent with major environmental and biogeographic gradients of the Western Palearctic. The weak positive temporal trend in core proportion is most plausibly attributable to sampling effects rather than to ecological processes, but the persistence of a well-defined core across records spanning nearly two centuries demonstrates the fundamental robustness of this spatial structure. Overall, our findings underscore the limitations of relying solely on extent-based metrics in conservation assessment, advocate for the routine use of point pattern approaches in macroecological studies of reptiles, and provide a methodological template for evaluating internal range structure in other widely distributed Western Palearctic species. By revealing how a single broadly distributed reptile species is internally organised at biogeographic scales, this work contributes to the broader understanding of

how Western Palearctic reptile biodiversity is spatially structured and how such structure can inform conservation prioritisation.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/d18060367/s1>. Table S1: complete list of the 3967 georeferenced occurrence records of *Pseudopus apodus* used in the analyses, including latitude, longitude, observation year, country, and data source.

Author Contributions: Conceptualisation, M.K.Ş. and M.K.; methodology, M.K.Ş. and M.K.; software, M.K.; validation, M.K.Ş., A.T. and M.K.; formal analysis, M.K.; investigation, M.K.Ş. and A.T.; data curation, M.K.Ş.; writing—original draft preparation, M.K.Ş. and M.K.; writing—review and editing, M.K.Ş., A.T. and M.K.; visualisation, M.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. This study did not involve direct manipulation of live animals; all analyses were based on previously published occurrence records obtained from publicly accessible biodiversity databases.

Data Availability Statement: The occurrence data analysed in this study are publicly available from the Global Biodiversity Information Facility (<https://www.gbif.org>) and iNaturalist (<https://www.inaturalist.org>). The compiled dataset and R code used for the analyses were deposited in Zenodo at: <https://doi.org/10.5281/zenodo.20555665> (accessed on 5 June 2025).

Acknowledgments: The authors thank the global community of contributors to GBIF, iNaturalist, and the *Atlas of European Reptiles*, whose records made this synthesis possible.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Brown, J.H. On the relationship between abundance and distribution of species. *Am. Nat.* **1984**, *124*, 255–279. [[CrossRef](#)]
2. Gaston, K.J. *The Structure and Dynamics of Geographic Ranges*; Oxford University Press: Oxford, UK, 2003.
3. Levin, S.A. The problem of pattern and scale in ecology. *Ecology* **1992**, *73*, 1943–1967. [[CrossRef](#)]
4. Wiens, J.A. Ecological heterogeneity: An ontogeny of concepts and approaches. *Oikos* **2000**, *91*, 451–457.
5. Sagarin, R.D.; Gaines, S.D. The ‘abundant centre’ distribution: To what extent is it a biogeographical rule? *Ecol. Lett.* **2002**, *5*, 137–147. [[CrossRef](#)]
6. Gaston, K.J. Geographic range limits: Achieving synthesis. *Proc. R. Soc. B* **2009**, *276*, 1395–1406. [[CrossRef](#)]
7. Holt, R.D. On the evolutionary ecology of species’ ranges. *Evol. Ecol. Res.* **2003**, *5*, 159–178.
8. Pironon, S.; Papuga, G.; Villellas, J.; Angert, A.L.; García, M.B.; Thompson, J.D. Geographic variation in genetic and demographic performance: New insights from an old biogeographical paradigm. *Biol. Rev.* **2017**, *92*, 1877–1909. [[CrossRef](#)]
9. IUCN. *Guidelines for Using the IUCN Red List Categories and Criteria; Version 16*; IUCN Standards and Petitions Committee: Gland, Switzerland, 2022.
10. Gaston, K.J.; Fuller, R.A. The sizes of species’ geographic ranges. *J. Appl. Ecol.* **2009**, *46*, 1–9. [[CrossRef](#)]
11. Channell, R.; Lomolino, M.V. Dynamic biogeography and conservation of endangered species. *Nature* **2000**, *403*, 84–86. [[CrossRef](#)]
12. Baddeley, A.; Rubak, E.; Turner, R. *Spatial Point Patterns: Methodology and Applications with R*; Chapman and Hall/CRC: Boca Raton, FL, USA, 2015.
13. Clark, P.J.; Evans, F.C. Distance to nearest neighbor as a measure of spatial relationships in populations. *Ecology* **1954**, *35*, 445–453. [[CrossRef](#)]
14. Ripley, B.D. Modelling spatial patterns. *J. R. Stat. Soc. B* **1977**, *39*, 172–212. [[CrossRef](#)]
15. Wand, M.P.; Jones, M.C. *Kernel Smoothing*; Chapman and Hall: London, UK, 1995.
16. Seaman, D.E.; Powell, R.A. An evaluation of the accuracy of kernel density estimators for home range analysis. *Ecology* **1996**, *77*, 2075–2085. [[CrossRef](#)]
17. Huey, R.B. Physiological consequences of habitat selection. *Am. Nat.* **1991**, *137*, S91–S115. [[CrossRef](#)] [[PubMed](#)]
18. Buckley, L.B.; Jetz, W. Environmental and historical constraints on global patterns of amphibian richness. *Proc. R. Soc. B* **2007**, *274*, 1167–1173. [[CrossRef](#)]

19. Sindaco, R.; Jeremčenko, V.K. *The Reptiles of the Western Palearctic. 1. Annotated Checklist and Distributional Atlas of the Turtles, Crocodiles, Amphisbaenians and Lizards of Europe, Caucasus and Asian Middle East*; Edizioni Belvedere: Latina, Italy, 2008.
20. Sillero, N.; Campos, J.; Bonardi, A.; Corti, C.; Creemers, R.; Crochet, P.A.; Crnobrnja-Isailović, J.; Denoël, M.; Ficetola, G.F.; Gonçalves, J.; et al. Updated distribution and biogeography of amphibians and reptiles of Europe. *Amphib.-Reptil.* **2014**, *35*, 1–31. [\[CrossRef\]](#)
21. Speybroeck, J.; Beukema, W.; Bok, B.; Van Der Voort, J. *Field Guide to the Amphibians and Reptiles of Britain and Europe*; Bloomsbury Publishing: London, UK, 2016.
22. IUCN. *Pseudopus apodus*. The IUCN Red List of Threatened Species. Available online: <https://www.iucnredlist.org/species/157248/746991> (accessed on 29 April 2026).
23. Chandler, M.; See, L.; Copas, K.; Bonde, A.M.Z.; López, B.C.; Danielsen, F.; Legind, J.K.; Masinde, S.; Miller-Rushing, A.J.; Newman, G.; et al. Contribution of citizen science towards international biodiversity monitoring. *Biol. Conserv.* **2017**, *213*, 280–294. [\[CrossRef\]](#)
24. Isaac, N.J.B.; van Strien, A.J.; August, T.A.; de Zeeuw, M.P.; Roy, D.B. Statistics for citizen science: Extracting signals of change from noisy ecological data. *Methods Ecol. Evol.* **2014**, *5*, 1052–1060. [\[CrossRef\]](#)
25. GBIF.org (27 October 2025) GBIF Occurrence Download. Available online: <https://doi.org/10.15468/dl.wdjfd2> (accessed on 29 April 2026).
26. Boria, R.A.; Olson, L.E.; Goodman, S.M.; Anderson, R.P. Spatial filtering to reduce sampling bias can improve the performance of ecological niche models. *Ecol. Modell.* **2014**, *275*, 73–77. [\[CrossRef\]](#)
27. R Core Team. *R: A Language and Environment for Statistical Computing*; R Foundation for Statistical Computing: Vienna, Austria, 2024.
28. Pebesma, E. Simple features for R: Standardized support for spatial vector data. *R J.* **2018**, *10*, 439–446. [\[CrossRef\]](#)
29. Wilcoxon, F. Individual comparisons by ranking methods. *Biom. Bull.* **1945**, *1*, 80–83. [\[CrossRef\]](#)
30. Wickham, H.; François, R.; Henry, L.; Müller, K.; Vaughan, D. *dplyr: A Grammar of Data Manipulation*, R Package Version 1.1.4; R Foundation for Statistical Computing: Vienna, Austria, 2023. Available online: <https://CRAN.R-project.org/package=dplyr> (accessed on 29 April 2026).
31. Golemund, G.; Wickham, H. Dates and times made easy with lubridate. *J. Stat. Softw.* **2011**, *40*, 1–25. [\[CrossRef\]](#)
32. Hahsler, M.; Piekenbrock, M.; Doran, D. dbSCAN: Fast density-based clustering with R. *J. Stat. Softw.* **2019**, *91*, 1–30. [\[CrossRef\]](#)
33. Aiello-Lammens, M.E.; Boria, R.A.; Radosavljevic, A.; Vilela, B.; Anderson, R.P. spThin: An R package for spatial thinning of species occurrence records for use in ecological niche models. *Ecography* **2015**, *38*, 541–545. [\[CrossRef\]](#)
34. Legendre, P.; Fortin, M.J. Spatial pattern and ecological analysis. *Vegetatio* **1989**, *80*, 107–138. [\[CrossRef\]](#)
35. Fortin, M.J.; Dale, M.R.T. *Spatial Analysis: A Guide for Ecologists*; Cambridge University Press: Cambridge, UK, 2005.
36. Keitt, T.H.; Bjørnstad, O.N.; Dixon, P.M.; Citron-Pousty, S. Accounting for spatial pattern when modeling organism–environment interactions. *Ecography* **2002**, *25*, 616–625. [\[CrossRef\]](#)
37. Hewitt, G.M. The genetic legacy of the Quaternary ice ages. *Nature* **2000**, *405*, 907–913. [\[CrossRef\]](#)
38. Pulliam, H.R. On the relationship between niche and distribution. *Ecol. Lett.* **2000**, *3*, 349–361. [\[CrossRef\]](#)
39. Turner, M.G. Landscape ecology: What is the state of the science? *Annu. Rev. Ecol. Evol. Syst.* **2005**, *36*, 319–344. [\[CrossRef\]](#)
40. Pulliam, H.R. Sources, sinks, and population regulation. *Am. Nat.* **1988**, *132*, 652–661. [\[CrossRef\]](#)
41. Hanski, I. Metapopulation dynamics. *Nature* **1998**, *396*, 41–49. [\[CrossRef\]](#)
42. Araújo, M.B.; Peterson, A.T. Uses and misuses of bioclimatic envelope modeling. *Ecology* **2012**, *93*, 1527–1539. [\[CrossRef\]](#)
43. Parmesan, C.; Yohe, G. A globally coherent fingerprint of climate change impacts across natural systems. *Nature* **2003**, *421*, 37–42. [\[CrossRef\]](#)
44. Chen, I.C.; Hill, J.K.; Ohlemüller, R.; Roy, D.B.; Thomas, C.D. Rapid range shifts of species associated with high levels of climate warming. *Science* **2011**, *333*, 1024–1026. [\[CrossRef\]](#)

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.